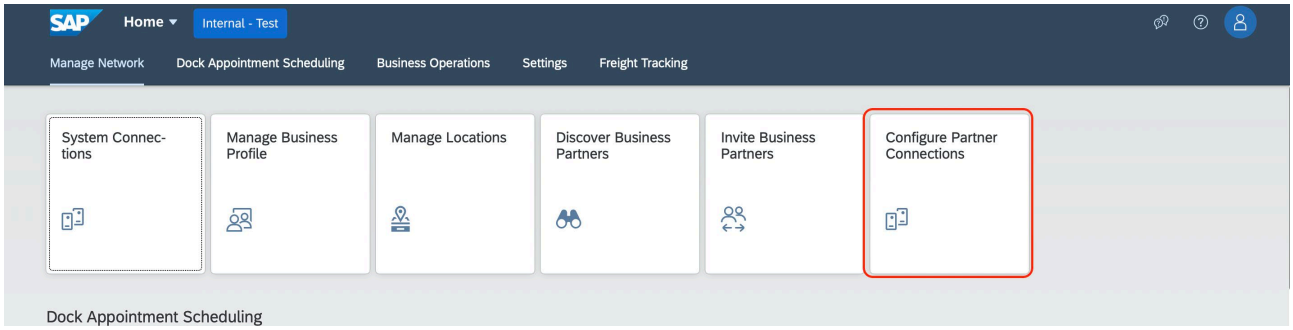
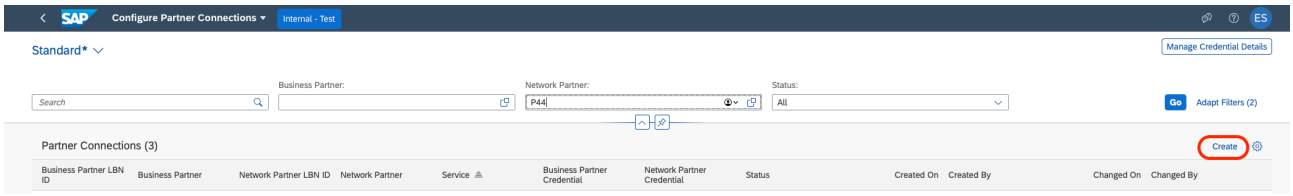
5. Configuring Connection Between Business Partners and project44

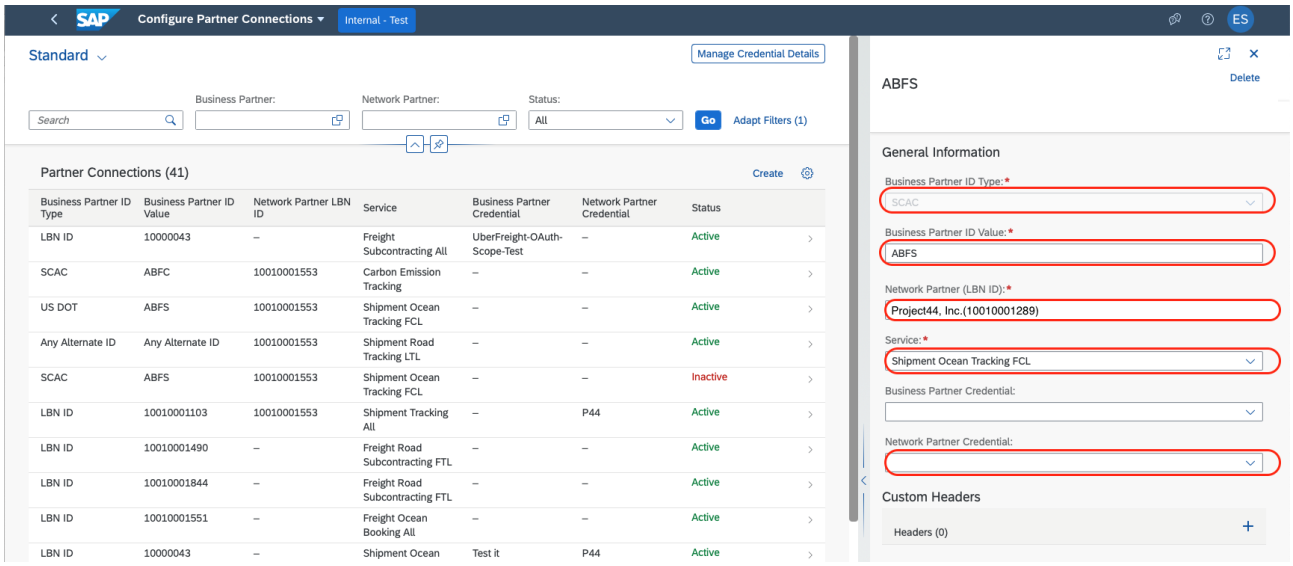
If all your business partners that use project44 services have LBN IDs, you are now ready to create a partner connection for each business partner.

Creating a Partner Connection Specific to Each Service for Each Business Partner

The partner connection with each business partner is created in the *Configure Partner Connections* app.







When filling out the Create New Partner Connection dialog, the Network Partner Credential is the project44 credential that was added as an OAuth network partner credential in Section 3 above. The Service is the project44 service with which the business partner is connected. And so, the partner connection in the Configure Partner Connections app is created unique to each service and for each business partner ID type and alternate ID. Thus, you map the business partner, project44, service, and credentials together to route calls from SAP Business Network for Logistics to project44.

In SAP Business Network Global Track and Trace, one LBN ID of project44 is only mapped with one of its alternate IDs. If project44 asks you to use another one for a particular business context, you can update the alternate ID as required by providing the key-value pair information under **Custom Headers**.

For more details, refer to Configure Partner Connections (freight collaboration) or Configure Partner Connections (global track and trace).

6. Configuration and Testing for SAP Business Network Freight Collaboration

6.1 Scenario Specific Configuration for SAP Business Network Freight Collaboration

With the B2B collaboration add-on, SAP Business Network enables the collaborative processing of FTL and LTL shipments with carriers.

Note:
For tracking-related scenarios, SAP recommends using SAP Business Network Global Track and Trace. To differentiate different scenarios, SAP Business Network for Logistics relies on the following fields:

- Transportation Mode – The transportation mode must be maintained in the freight order document (SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA)
- Shipping type –The shipping type must be maintained in the freight order document (SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA)

Based on the scenario, additional data is required from the SAP Transportation Management (SAP TM) system:

a. *Road FTL Tracking*
b. *Road FTL Tendering & Subcontracting*
c. *Road LTL Subcontracting*
d. *Ocean tracking*
    i. project44 has to mark the webhook as the default for ocean tracking scenarios. This has to be done one time with an email request to project44 [implementation@project44.com]

FTL Tendering and Subcontracting Scenario

As the first step in the tendering process, you – as an ordering party – create a request for freight quotation (RFQ) in your SAP TM system. When you publish this RFQ, the system sends the RFQ to SAP Business Network for Logistics. SAP Business Network for Logistics uses the data in the RFQ and creates a tendered shipment and a tender in project44 with the details of the origin, destination, truck type, and so on.

The system creates the tender and tendered shipment. You can view the tendered shipment ID and tender ID in the document reference details of the RFQ.

Note that the carrier for which you have created the RFQ must be a project44 carrier. This carrier can view this tender and send one of the following responses:

- **Accept**
- **Accept with Rate Quote**
- **Reject**

Once the carrier responds to the tender from the project44 system, SAP Business Network for Logistics creates a quotation based on this response and sends it to your SAP TM system. You can either award or cancel the quotation.

When you accept the response, the SAP TM system sends a freight order for confirmation to the carrier in SAP Business Network for Logistics for subcontracting which in turn triggers a dispatch in the project44 system. When you send a cancellation, the system cancels the dispatch in the project44 system.

When you cancel the RFQ, the system updates the status to cancelled in SAP Business Network for Logistics and deletes the corresponding tendered shipment and tender in project44.

Response Required Scenario

In your SAP TM system, you can control whether you require a response from your carrier before the carrier starts the dispatch. If you do not require a response from the carrier, the carrier begins the dispatch directly after the system creates an RFQ. If you require a response, then the carrier triggers the dispatch only after the carrier receives the freight order for confirmation.

LTL Subcontracting Scenario

In the LTL scenario, you have precontracted rates with the carrier. Hence the carrier starts the dispatch directly without any RFQs.

Once you send the freight order from your SAP TM system, the system creates a shipment in project44, and the carrier begins the dispatch. Note that if you cancel the freight order in your SAP TM system, using the shipment ID, the system cancels the dispatched shipment in project44.

SAP Business Network, project44 FTL visibility add-on

All freight orders created in your SAP TM or SAP S/4HANA system can be tracked with the project44 add-on for FTL visibility. You must ensure that the carrier is assigned to the freight order and that the carrier is available in the project44 network. When the freight order is set to *In Execution*, it is sent to SAP Business Network for Logistics, and you can display it in the *Track Freight Movement* app.

To use FTL tracking in SAP Business Network for Logistics, you must provide the following information in your SAP TM or SAP S/4HANA system when creating the freight order:

- On the *Stages* tab:
  Source Location and *Destination Location*
  Also, either the *Planned Departure Time* or *Planned Arrival Time* must be in the future, and the arrival time must be later than the departure time.
- On the *Terms and Conditions* tab: *Shipping Type* must be FTL (18).
- On the *General Data* tab: *Registration Country/No*.

Note: It is now possible in LBN for shippers to maintain a default shipping type in the *Assign Codes* app. So, when no shipping type is maintained for a shipment, the default value will be used, assuming that one has been maintained.

The following events are supported for FTL:

- Departure
- Delay
- Arrival at Destination
- Expected Time of Arrival

The reported events are sent to your SAP TM or SAP S/4HANA system and are visible on the *Execution* tab of the freight order.

SAP Business Network, project44 LTL visibility add-on

All freight orders created in your SAP TM or SAP S/4HANA system can be tracked with the project44 add-on for LTL visibility. You must ensure that the carrier is assigned to the freight order and that the carrier is available in the project44 network. When the freight order is set to *In Execution,* it is sent to SAP Business Network for Logistics, and you can display it in the *Track Freight Movement* app.

To use LTL tracking in SAP Business Network for Logistics, you must provide the following information in your SAP TM or SAP S/4HANA system when creating the freight order:

- **Under the *Stages* tab:**
  Source Location and *Destination Location*.
  The arrival time must be further in the future than the departure time. It is only possible for the departure time to be in the past. The arrival time, at least, must be in the future.
  Only one stage (with source location and destination location) is supported within the LTL process. Multiple stages are not supported.
- **Under the *Terms and Conditions* tab:**
  Shipping Type must be LTL (17).
- **Under the *Document References* tab:**
  Tracking can only start when the freight order that's sent to the carrier includes the following in the *Document Types for Business Transaction* field:
  - PRO number: 'BN' (Consignment identifier, carrier assigned)
  - Bill of lading: 'T50' (Bill of Lading)

Note: It is now possible in LBN for shippers to maintain a default shipping type in the *Assign Codes* app. So, when no shipping type is maintained for a shipment, the default value will be used, assuming that one has been maintained.

The following events are supported for LTL:

- Departure
- Delay
- Arrival at Destination

The reported events are sent to your SAP TM or SAP S/4HANA system and are visible under the *Execution* tab of the freight order. SAP Business Network for Logistics receives the attachment related to an event from project44 as a URL for proof of pickup and proof of delivery. This URL will be accessible in the *Track Freight Movement* app.

Additionally, the carrier can upload the attachments in the *Freight Orders for Reporting* app.

SAP Business Network, project44 ocean visibility add-on

All ocean bookings created in your SAP TM or SAP S/4HANA system can be tracked with the project44 add-on for ocean visibility. You must ensure that the carrier is assigned to the ocean booking and that the carrier is available in the project44 network. When the ocean booking is set to *In Execution*, it is sent to SAP Business Network for Logistics, and you can display it in the *Track Freight Movements* app.

To use ocean tracking in SAP Business Network for Logistics, you must provide the following information in your SAP TM or SAP S/4HANA system when creating the freight order:

If you want your carriers to upload attachments (proof of pick-up, proof of delivery) as part of the reporting, they can do so in the *Freight Orders for Reporting* app.

- Under the *Stages* tab:
  Source Location and *Destination Location*.
  The locations should mandatorily have a UN/LOCODE maintained. This is later used to correlate the events reported by project44 onto TM planned events.
  Only one stage (with source location and destination location) is supported. Multiple stages are not supported.
- Under the *Terms and Conditions* tab:
  Shipping Type must be FCL (full container load) or LCL (less than container load).
- Under the *Notes* tab:
  A note with text type 'Container ID' must be created with the container ID of the ocean booking as the value. This is a custom text type. Note that you must separate multiple container IDs using commas.

Note: It is now possible in LBN for shippers to maintain a default shipping type in the *Assign Codes* app. So, when no shipping type is maintained for a shipment, the default value will be used, assuming that one has been maintained.

The system supports multiple events for ocean tracking. The following are some of the important events:

- Departure
- Arrival at Destination
- Gate-In End
- Gate-Out End
- Loading End
- Unloading End
- Return

The reported events are sent to your SAP TM or SAP S/4HANA system and are visible on the *Execution* tab of the freight order. If you want your carriers to upload attachments (proof of pickup, proof of delivery) as part of the reporting, they can do so in the *Freight Orders for Reporting* app.

6.2 Testing in SAP Business Network Freight Collaboration

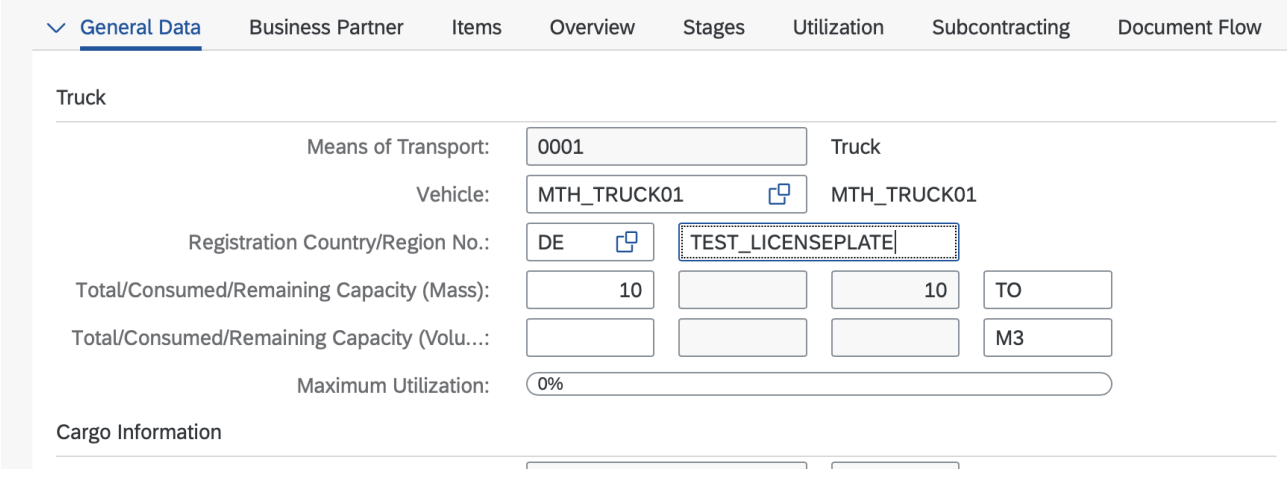
Tracking Initialization – flow from SAP Business Network for Logistics to project44:
Trigger a freight order out from TM system by setting Execution status or from ECC. The order will be pushed to the project44 system. Successful orders sent can be verified by logging in to project44. The URL to log in to project44 (https://na12.api.project44.com) would likely have been maintained by you in the *Configure Partner Connections* app.

Event flow from project44:
You may get events from the project44 system based on your requirements. Simulated events are supported only for Road FTL. For other scenarios, the testing has to be done manually with project44.

Road FTL:
The simulation of events must be requested from SAP with an incident categorized by component SBN-LBN-INT. To create an incident, open https://launchpad.support.sap.com/#incident/create. And this option is available only in the shipper test tenant.

To test with real road orders, you might have to obtain the vehicle ID, license plate (truck) or driver mobile no. from project44 which are relevant for your carrier executing the orders. As mentioned above, these vehicle identifiers have to be maintained in the SAP TM or ECC system. This information has to flow from your system to SAP Business Network for Logistics to project44 during tracking initialization. When the initialization of tracking is done in the p44 system, the events will flow to your system. To monitor if the messages are being processed from SAP Business Network for Logistics as expected, you may view them in the *Manage Integration Logs* app (freight collaboration) or in the *Track Freight Movement* app (freight collaboration).

Example of how to maintain Shipment in TM to maintain these identifiers:



Road LTL:
Simulation currently not available
You must obtain a real (not test) Shipment identifier (T50 – Bill of lading, T67- Pro Number) from project44 and maintain it in the SAP TM system in the document reference section of the freight

order or shipment. This information must be sent during initialization. After project44 receives it and when the actual event occurs, the events from project44 will be sent which can be validated using the Manage Integration Logs app.

Example of how to maintain Shipment in SAP TM to maintain these identifiers:

Document References

Insert

Business Transaction Document ID	Document Type for Business Transaction	Document Type for Business Transaction (Desc.)	Item
1570866519628	825	Dispute Case	
1570422048103	825	Dispute Case	
1574394404492	825	Dispute Case	
BOL_TEST	T50	Bill of Lading	
PRO_TEST	T67	Reference number of Carrier	

Ocean:

Simulation currently not available

You must obtain the real container ID (CONTAINER_ID) or Bill of lading (T50) from project44 and maintain it in TM freight order document reference or in global track and trace. This must be sent during track initialization. Based on the real ocean events, the events will be reported from project44. To monitor if the messages are being processed from SAP Business Network for Logistics as expected, you may view them in the *Manage Integration Logs* app (freight collaboration) or in the *Track Freight Movement* app (freight collaboration).